

STA623 - Bayesian Data Analysis

Practical 1: Bayesian reasoning: prior, likelihood, posterior

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Goals. Use Bayes' rule numerically; compute posteriors by grid approximation for one-parameter models; derive a conjugate posterior and check grid results against it; see how prior and sample size interact.

Exercise 1: combining two diagnostic tests

A new diagnostic test for a disease D has sensitivity $P(T \mid D) = 0.99$ and specificity $P(\bar{T} \mid \bar{D}) = 0.90$. An existing test has sensitivity 0.80 and specificity 0.95. Given disease status, the two tests are independent.

- (a) With prevalence 10%, compute $P(D \mid T)$ for each test, and $P(D \mid \text{both positive})$. *Hint: define a compound test that is positive only if both tests are positive.*
- (b) Write an R function `ppv(prev, sens, spec)` and plot $P(D \mid T+)$ against prevalence (0.1% to 50%) for the new test, the old test and the compound test on one graph. At which prevalence does the new test's positive predictive value reach 50%?
- (c) Sequential testing: a patient tests positive on the new test. Use the posterior from that result as the *prior* for the old test. Show that this gives the same answer as the compound test in (a). Why?

Exercise 2: grid approximation for a prevalence

A school survey tests $n = 40$ children for *Schistosoma haematobium*; $k = 11$ are positive. Earlier surveys in the district suggest a prevalence around 30%, and you encode this as a $\text{Beta}(3, 7)$ prior.

- (a) **Exact posterior.** Assuming $k \mid \pi \sim \text{Bin}(n, \pi)$ and the prior $\pi \sim \text{Beta}(a, b)$, derive the posterior distribution $p(\pi \mid k)$ analytically and state its parameters for these data. Then use R (`qbeta`, `pbeta`) to compute the exact posterior mean, median, 95% equal-tailed credible interval and $P(\pi > 0.35 \mid k)$.
- (b) Compute the posterior of the prevalence π on a grid of 1001 points. Plot prior, (scaled) likelihood and posterior.

- (c) Draw 10^5 samples from the grid posterior and use them to estimate the posterior mean, median, a 95% credible interval and $P(\pi > 0.35 \mid k)$.
- (d) Compare with the exact posterior summaries from (a).
- (e) Repeat (b) using a grid of only 11 points. What goes wrong?

Exercise 3: a discrete prior

A bag contains six coins: three with $P(\text{heads}) = 0.8$, two with 0.3 and one fair coin. One coin is drawn at random and tossed $n = 6$ times, giving $k = 2$ heads.

- (a) Compute the posterior probability of each coin type. (This is a grid with three points.)
- (b) The same coin is tossed another 6 times, giving 5 heads. Update your posterior. Then check that analysing all 12 tosses at once gives the same result.
- (c) Can this analysis ever conclude that $\pi = 0.6$? Comment.

Exercise 4: grid approximation for a Poisson rate, conjugate and non-conjugate

Number of malaria episodes in one season for 18 children:

```
y <- c(1, 3, 2, 3, 0, 2, 6, 4, 4, 1, 1, 3, 2, 3, 1, 1, 3, 0)
```

Model: $Y_i \mid \lambda \sim_{\text{iid}} \text{Pois}(\lambda)$, with prior $\lambda \sim \text{Gamma}(a, b)$, $a = 5$, $b = 2$ (shape a , rate b).

- (a) **Exact posterior.** Derive the posterior distribution $p(\lambda \mid \mathbf{y})$ analytically. *Hint: write down the likelihood and the prior, keep only the factors that depend on λ , and recognise the kernel of a known distribution.* State its parameters for these data, and use R to compute the exact posterior mean and 95% equal-tailed credible interval. Through which summary of the data does $\mathbf{y}$ enter the posterior?
- (b) **Grid approximation.** Compute the posterior of λ on a grid over $(0, 6)$, working on the **log scale** (compute log-likelihood + log-prior, subtract the maximum, exponentiate, then normalise by dividing by the sum over the grid). Plot it together with your exact posterior from (a), and compare the posterior mean and 95% credible interval obtained from the grid with the exact values.
- (c) Now use a prior that is *not* conjugate: a half-normal, $\lambda \sim \mathcal{N}^+(0, 3^2)$ (use `2 * dnorm(lambda, 0, 3)`). Can you still derive the posterior in closed form? Compute the grid posterior, its mean and 95% credible interval. Does the change of prior matter?
- (d) Why is it important to work on the log scale? Try computing `prod(dpois(rep(y, 50), 2))` (i.e. 50 times as much data).

Exercise 5: how much does the prior matter?

For the prevalence setting of Exercise 2, suppose the observed proportion stays at 27.5% while the sample size is $n = 10, 40, 200, 1000$.

- (a) For each n , compute the posterior mean and 95% credible interval under three priors: Beta(1, 1), Beta(3, 7) and a strongly informative but *wrong* Beta(60, 40) (prior mean 60%).
- (b) Plot the results. Describe when the prior matters, and how the “prior sample size” $a + b$ helps you predict this.

Exercise 6

Suppose $\Pi \sim \text{Beta}(a, b)$ and $Y \mid \pi \sim \text{Bin}(n, \pi)$. Show that

$$E[\Pi \mid Y = k] = \frac{a + b}{a + b + n} E[\Pi] + \frac{n}{a + b + n} \frac{k}{n}.$$

What happens as $n \rightarrow \infty$?